Harish Manoharan

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Software engineer at Purgo AI working on agents, retrieval and LLM evaluation. M.S. Computer Science, UT Dallas, 2026 (GPA 3.92/4.00); dual degree in Data Science and Biological Sciences, IIT Madras.

EXPERIENCE

Purgo AI San Francisco Bay Area (Los Altos, CA)

AI startup building an agent that writes and validates Databricks and dbt code

Software Engineer Jun 2026 – Present

Software Engineer Intern (remote) Jun 2025 – May 2026

- Cut missed source tables from 79 to 24 on our 103-ticket internal benchmark by adding dbt lineage expansion to the LangGraph agent's retrieval (sqlglot parses the model a ticket names and follows its ref() and source() calls); picked the design by ablation and rejected a repo-search alternative that added 4.7x more unwanted tables.
- Built Purgo's IQ/OQ validation engine (NestJS/TypeScript, GCP Cloud Run), which qualifies life-sciences (GxP) customers' Databricks platforms on Azure and AWS: it runs a git-versioned library of 59 tests as cancellable jobs with live progress, keeps redacted evidence for signed audit reports, and turns requirement documents into new tests through an LLM pipeline with reviewer feedback.
- Built the LLM validation product across UI, API and test engine (React/Next.js, NestJS, PostgreSQL): 19 tests that score a customer's Azure OpenAI or Databricks-served model for hallucination on missing fields, per-field extraction F1, demographic bias, determinism over 20 runs and latency at 20 concurrent calls, plus checks on the deployment's access, network and audit setup.
- Turned recurring Databricks runtime failures into AST lint rules that the agent's self-review runs on generated PySpark/SQL, so those errors surface in review and runtime failures dropped; added dbt source checks and a recovery pass that keeps the candidate passing the most checks when every one fails.
- Built semantic search over Polaris/Iceberg catalogs on Pinecone (incremental re-indexing, keyword fallback) and its 188-query benchmark: recall@10 of 0.68 vs 0.54 and mean latency of 0.18 s vs 4.0 s for the existing search.
- Rewrote attachment chunking to keep every record whole and tell the model which parts it didn't get; moved two agent stages to a smaller reasoning model after five 102-ticket benchmark runs showed recall within run-to-run noise (7.6x cheaper on those stages).

Digitus Business Solutions Bengaluru, India

Software Engineer Intern May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building a text-to-SQL agent with schema discovery and retries driven by execution errors.
- Built a multimodal RAG assistant over internal slide decks and videos, with layout-aware chunking and source citations.

PwC India Bengaluru, India

Technology Consulting Intern, Data Analytics May 2023 – Jul 2023

- Cut forecast error (MAPE) 15% against the baseline across 50+ SKUs with an XGBoost + SARIMA ensemble.

EDUCATION

The University of Texas at Dallas M.S. Computer Science May 2026

GPA 3.92/4.00 · Dean's Graduate Scholar

Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences May 2024

INSPIRE Scholar · Young Research Fellow · Thesis with The Jackson Laboratory on unsupervised behavior discovery in mice

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Machine learning and LLMs: retrieval-augmented generation (RAG), LLM evaluation, AI agents (LangGraph), PyTorch

Data: Databricks, Apache Spark (PySpark), dbt, PostgreSQL, Apache Iceberg

Web, backend and cloud: React, Next.js, FastAPI, NestJS, REST APIs, AWS, Azure, GCP, Docker, CI/CD (GitHub Actions)